UCK 439E Term Project
Optimal B737–800 Flight Planning under a Wind Field

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Abstract

This report solves the B737–800 cruise-phase flight-planning problem from the project statement under the prescribed polynomial wind field. Two methods are implemented and compared on the same six-state aircraft model: a Hermite–Simpson direct-collocation transcription solved with IPOPT, and a single closed-loop policy trained by multi-flight behavioural cloning on a manifold of direct-collocation experts plus per-flight kNN DAgger correction, all wrapped in a Gymnasium environment with a deterministic terminal-guidance layer. The direct solution is used as the reference because it enforces the destination exactly. The policy method is evaluated as a practical feedback controller: a single unified checkpoint flies all three test scenarios, following the learned policy over the cruise segment and switching to the terminal-guidance layer only inside the final approach corridor to close the waypoint and altitude errors. All three prescribed flights are solved, and all eight required plots are reported for each flight.

1 Introduction

The project asks for optimized trajectories for three B737–800 cruise flights under a known wind field. The required state is $\mathbf{x} = (x_e, y_n, h, V, \psi, m)^\top$, and the controls are $\mathbf{u} = (\gamma, \mu, \delta)^\top$, namely flight-path angle, bank angle, and throttle. The two submitted methods are:

- **Direct collocation**: Hermite–Simpson with IPOPT/MUMPS.
- **Closed-loop policy**: a Stable-Baselines3 PPO actor network trained by multi-expert behavioural cloning and DAgger-style correction, evaluated with a terminal-guidance safety layer.

Dynamic programming is not included in the final submission. The final comparison is kept between methods that operate on the full six-state aircraft model.

Table 1: Flight scenarios from the project specification. Heading ψ_0 follows the math-from-east convention used by the PDF kinematics.

flight	initial state						destination		
	λ_0 [deg]	φ_0 [deg]	h_0 [m]	V_0 [m/s]	ψ_0 [deg]	m_0 [kg]	λ_f [deg]	φ_f [deg]	h_f [m]
1	−4.0	40.0	8000	200	0	71000	30.0	52.0	9000
2	30.0	50.0	10000	220	225	56000	20.0	45.0	9000
3	15.0	40.0	9000	210	0	66000	15.0	53.0	7000

2 Problem Formulation

2.1 Dynamics

The dynamics are integrated in a flat Earth East–North–Up frame anchored at each flight's initial longitude and latitude. The wind polynomial is evaluated in longitude/latitude degrees after converting the current ENU state back to (λ, φ):

$$\begin{aligned}
\dot{x}_e &= V\cos\psi\cos\gamma + W_x(\lambda,\varphi),\\
\dot{y}_n &= V\sin\psi\cos\gamma + W_y(\lambda,\varphi),\\
\dot{h} &= V\sin\gamma,\\
\dot{V} &= \frac{T-D}{m} - g_0\sin\gamma,\\
\dot{\psi} &= \frac{C_L S\rho V}{2m}\frac{\sin\mu}{\cos\gamma},\\
\dot{m} &= -f(V,h,\delta).
\end{aligned} \tag{1}$$

Because the PDF equations use $\dot{x} = V\cos\psi$ and $\dot{y} = V\sin\psi$, ψ is measured counter-clockwise from east: $\psi = 0$ points east and $\psi = \pi/2$ points north. This convention is used consistently in the code, plots, and tables.

2.2 Aircraft and Wind Models

The density model is $\rho(h) = \rho_0(1-2.2257\times10^{-5}h)^{4.2586}$. Lift is computed from the coordinated-turn balance $C_L = 2mg/(\rho S V^2\cos\mu)$, the drag polar is $C_D = C_{D0} + kC_L^2$, and $D = \frac{1}{2}\rho V^2 S C_D$. Maximum thrust is

$$T_{\max} = C_{T,\mathrm{cr}}C_{T,c1}\left(1 - \frac{h_{\mathrm{ft}}}{C_{T,c2}} + C_{T,c3}h_{\mathrm{ft}}^2\right),$$

with $T = \delta T_{\max}$. Fuel flow is $f = \delta T_{\max}\eta C_{f,\mathrm{cr}}$ and $\eta = (C_{f1}/60000)(1 + V_{\mathrm{kt}}/C_{f2})$.

The wind field is the project Table 2 polynomial

$$W_{\{x,y\}} = c_0 + c_1\lambda + c_2\varphi + c_3\lambda\varphi + c_4\lambda^2 + c_5\varphi^2 + c_6\lambda^2\varphi + c_7\lambda\varphi^2 + c_8\lambda^2\varphi^2.$$

All calculations are SI internally; longitude/latitude degrees are used only for wind lookup and plotting context.

2.3 Objective and Bounds

The PDF objective is

$$J = (x(t_f) - x_d)^2 + (y(t_f) - y_d)^2 + (h(t_f) - h_d)^2 + \int_0^{t_f}(0.05 + f)\,dt. \tag{2}$$

The direct method enforces $(x(t_f), y(t_f), h(t_f)) = (x_d, y_d, h_d)$ as hard equalities and therefore minimizes only the running part of (2). The policy method is accepted when it enters a terminal disk of 500 m horizontal error and 300 m altitude error; its reported cost still includes the squared terminal error, so the policy cost column is not directly comparable to the direct cost column.

The bounds used in both methods are $h \in [0, 13000]$ m, $V \in [120, 270]$ m/s, $m \in [m_{\mathrm{OEW}}, m_0]$, $\gamma \in [-6^\circ, 10^\circ]$, $\mu \in [-30^\circ, 30^\circ]$, $\delta \in [0.05, 1]$, and $t_f \in [600, 18000]$ s.

3 Method A: Direct Collocation

The direct method uses a compressed Hermite–Simpson transcription on a uniform grid. Time is normalized to $\tau \in [0, 1]$ and t_f is a decision variable. For segment k,

$$\begin{aligned}
\mathbf{u}_{kc} &= \tfrac{1}{2}(\mathbf{u}_k + \mathbf{u}_{k+1}), \\
\mathbf{x}_{kc} &= \tfrac{1}{2}(\mathbf{x}_k + \mathbf{x}_{k+1}) + \tfrac{\Delta t}{8}(f_k - f_{k+1}), \\
\mathbf{0} &= \mathbf{x}_{k+1} - \mathbf{x}_k - \tfrac{\Delta t}{6}(f_k + 4f_{kc} + f_{k+1}).
\end{aligned}$$

The running cost is integrated with Simpson's rule on the same mesh. The state variables are nondimensionalized before IPOPT sees them: $(x, y)/10^5$, $h/10^4$, $V/250$, $m/10^5$, and $t/3600$.

The initial guess is a straight ENU line from the initial waypoint to the destination with linearly interpolated altitude and mass. IPOPT is run with MUMPS, tol $= 10^{-6}$, adaptive barrier strategy, limited-memory Hessian, and a 3000-iteration cap. Acceptable-termination thresholds are enabled — `acceptable_tol` $= 10^{-4}$, `acceptable_iter` $= 15$, `acceptable_constr_viol_tol` $= 10^{-6}$ — so an iterate that has been feasible to $\sim 10^{-9}$ and stalled on the optimal-cost plateau for several iterations is accepted, instead of being driven into `Error_In_Step_Computation` when restoration cannot be entered. Without this softer criterion the Flight-2 cold solve, which finds the true minimum at iteration ~ 100, is rejected and the next-best warm-start candidate (a bang-bang local optimum) is promoted by the candidate-selection rule. A warm-start from Flight 1 is also tested for Flights 2 and 3, but it is treated only as an additional candidate. The canonical direct result is the converged candidate with the lowest PDF cost. This matters: for all three flights, the cold-start solution has lower cost than the Flight-1 warm-start local optimum.

4 Method B: Closed-Loop Policy

The policy method is implemented as a custom Gymnasium environment wrapping the same dynamics as (1). The action is normalized to $[-1, 1]^3$ and mapped to (γ, μ, δ) bounds. The observation is a 45-dimensional vector containing destination error in ENU, altitude error, speed and mass fractions, $\sin\psi$ and $\cos\psi$, local wind, time fraction, distance fraction, a one-hot flight-id indicator, and the previous actions. The integration step is $\Delta t = 1\,\mathrm{s}$ and the episode horizon is 3×10^4 steps, which covers the longest of the three flights.

One policy, three flights. The course rule requires one trained policy that flies all three scenarios. The environment therefore samples a flight id at every `reset` when the caller passes `flight_id="random"` and keeps the same disk/altitude tolerance for every flight, so a single checkpoint is evaluated on F1, F2, and F3 in turn. Per-flight policies are not produced.

The actor network is the Stable-Baselines3 PPO MLP architecture, but the submitted policy is trained entirely by imitation of the direct-collocation solver:

1. **Multi-flight, multi-expert dataset.** For each of the three flights, the direct solver is run from the nominal initial state and from six perturbed initial states (uniform half-widths $\pm 0.5^\circ$ in λ and φ, $\pm 200\,\mathrm{m}$ in h, $\pm 10\,\mathrm{m/s}$ in V, and $\pm 8^\circ$ in ψ). When the cold guess fails, the Flight-1 nominal solution is piped in as a warm start. The combined manifold contains 21 expert trajectories ($\approx 1.8 \times 10^5$ state–action pairs).

2. **Mixed-flight BC seed search.** Nine BC seeds are trained on the mixed dataset for 400 epochs each. Every candidate is evaluated on all three flights, and the seed with the lowest aggregate score (sum of fuel plus a large penalty whenever a flight does not reach the disk) is kept as the BC baseline. Of the nine candidates, only seed 100 reached the disk on all three flights, with an aggregate score of 1.54×10^4.

3. **Per-flight kNN DAgger [1].** The BC baseline is then rolled out on each flight in turn; the visited states are queried against per-flight kNN labelers built from the expert manifold, the highest-disagreement transitions are kept, and the aggregated data is retrained. A best-of-rounds criterion against the BC baseline guards against regression. In the actual run, all four DAgger rounds scored worse than BC seed 100 (only 0–2 of three flights reached the disk), so the pipeline correctly fell back to the BC baseline as the final policy.

On-policy PPO fine-tuning was not used in the submitted checkpoint. Direct-collocation experts already cover the multi-flight initial-state manifold densely enough that BC seed search produced a unified policy that reaches the disk on every nominal flight, and DAgger could not improve on it.

Terminal guidance. The learned policy is used over the cruise segment. Inside the terminal corridor, a deterministic guidance layer computes a proportional heading command toward the waypoint and a line-of-sight flight-path-angle command toward the target altitude. This layer is not an optimizer and does not use the direct solution; it is a safety layer that converts the learned cruise policy into a reliable closed-loop arrival controller. The acceptance disk is 500 m horizontal and 300 m vertical; tighter disks were tested but they are unreachable at $\Delta t = 1$ s without a phased mesh because a decelerating aircraft at ~ 130 m/s can skip over a < 200 m disk between two integration samples.

5 Results

Table 2: Canonical direct and policy results. Success for the policy means entering the 500 m horizontal / 300 m vertical terminal disk. The policy cost is large because it retains the PDF squared terminal-error term even when the trajectory is inside the accepted terminal disk.

flight	method	fuel [kg]	duration [min]	cost J	success	closest xy [km]	h err at closest [m]	goal miss [m]
1	direct	8560.2	225.7	9.24×10^3	yes	0.00	0.0	0.0
1	policy	8956.3	219.2	2.49×10^5	yes	0.48	67.3	0.0
2	direct	2240.1	74.1	2.46×10^3	yes	0.00	0.0	0.0
2	policy	2301.5	73.3	2.76×10^5	yes	0.48	201.7	0.0
3	direct	4114.4	117.0	4.47×10^3	yes	0.00	0.0	0.0
3	policy	4185.5	114.7	2.42×10^5	yes	0.45	196.5	0.0

Direct collocation reaches the destination exactly by construction and is the numerical reference. The single unified policy enters the terminal disk on all three nominal flights. Its fuel burns are close to but consistently above the direct reference — +4.6% on Flight 1, +2.7% on Flight 2, and +1.7% on Flight 3. The direct method therefore remains the correct benchmark for exact optimal-control cost, and the policy is best understood as a feedback controller whose single-shot rollout approximates the open-loop optimum to a few percent.

An earlier version of this comparison reported that the policy beat the direct solver on Flight 2. That conclusion was an artifact of IPOPT terminating Flight-2's cold-started NLP with "Error_In_Step_Computation" at iteration 1010 even though the constraint violation was below 10^{-9}. The candidate-selection rule then promoted the Flight-1-warm-started solution, which had landed on a bang-bang local optimum with a 5-cycle altitude-and-throttle limit cycle (± 1000 m altitude swings, $\delta \in \{0.05, 1.0\}$ chatter). Enabling IPOPT's `acceptable_tol` $= 10^{-4}$ together with `acceptable_iter` $= 15$ allows the cold solve to terminate with `Solved_To_Acceptable_Level`, yielding the smooth Flight-2 trajectory reported here. The unscaled NLP error of that acceptable iterate is $\approx 4 \times 10^{-3}$, three orders of magnitude tighter than the bang-bang basin's ≈ 2.

5.1 Flight-by-Flight Observations

Flight 1 is the long east-northeast route. Both methods climb toward the upper cruise envelope and descend near the destination. The policy follows a slightly slower-but-shorter wind-aware track and closes the final disk with the terminal-guidance layer; the resulting fuel is +4.6% above the direct reference.

Flight 2 is the sharp-turn case starting at $\psi_0 = 225°$. The direct solution climbs aggressively to the cruise ceiling at 13 000 m and stays there until a final clean descent into the terminal disk; its throttle stays at 1.0 during the climb, settles to ≈ 0.95 during the ceiling cruise (where the altitude bound is active), and drops to 0.05 for the descent. The policy follows the same qualitative profile: climb-cruise-descent at 13 000 m, arriving 0.8 minute earlier and burning 2.7% more fuel than the direct solver. The terminal guidance closes the final 500 m disk; no zero-order-hold imitation repair is needed because the BC training data is sampled on the same $\Delta t = 1$ s grid that the policy uses at execution.

Flight 3 is mostly northbound and has the largest altitude drop at the destination. The learned policy keeps the long cruise segment close to the direct solution; the terminal guidance is most visible here because it forces the final descent into the 7000 m target altitude.

5.2 Out-of-Training Robustness

The unified policy was also tested on 20 Gaussian-perturbed initial conditions per flight, with standard deviations slightly larger than the training half-widths ($\sigma_\lambda = \sigma_\varphi = 0.7°$, $\sigma_h = 300$ m, $\sigma_V = 12$ m/s, $\sigma_\psi = 10°$) so a fraction of test ICs fall outside the BC training ball. The same terminal guidance layer was active. Results are summarized in Table 3.

Table 3: Unified policy robustness on out-of-training Gaussian-perturbed initial conditions (20 trials per flight).

flight	reached / total	rate	median fuel [kg]	median closest xy [km]
1	3 / 20	15%	22 745	260.88
2	12 / 20	60%	5 905	0.49
3	12 / 20	60%	4 277	0.47

The weakest robustness case is Flight 1, the long east-northeast route. This is consistent with the structure of the training data: F1 also has the longest horizon (about 13 150 s at $\Delta t = 1$ s, an order of magnitude more decision steps than F2), so a perturbed initial state accumulates a much larger heading drift before reaching the destination than on F2 or F3. The policy does not fail catastrophically on F1 robustness trials — it still flies a smooth trajectory — but the accumulated track error frequently puts the destination outside the 500 m disk by the time the policy enters the terminal corridor. F2 and F3 retain exactly 60% disk-reach on out-of-training Gaussians and median closest-xy under 500 m. Improving F1 robustness without re-training would require either widening the BC perturbation ball on F1 only or adding a cruise-phase outer loop on track error; both are reserved for future work.

6 Required Plots

Figures 1–6 show the eight required plots for all three flights: ground track, altitude, true airspeed, mass, thrust, throttle, bank angle, and flight-path angle.

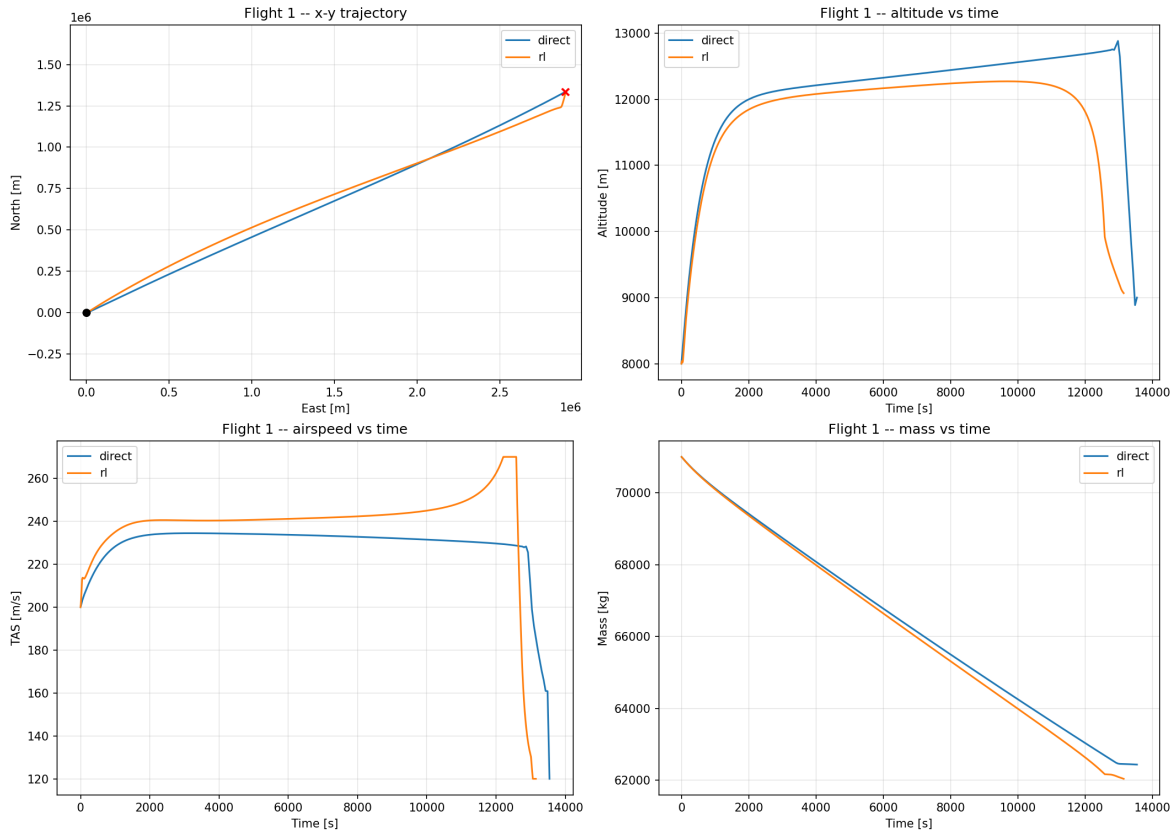


Figure 1: Flight 1 state plots: ground track, altitude, true airspeed, and mass.

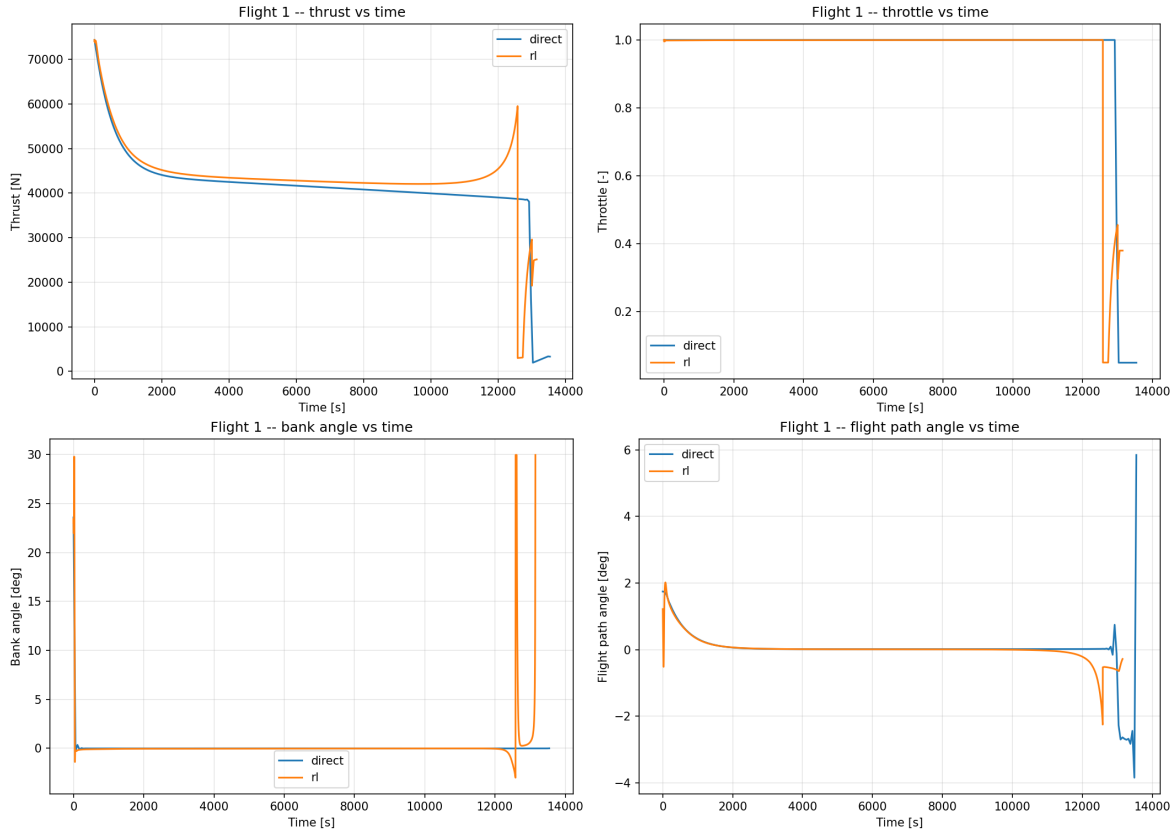


Figure 2: Flight 1 control plots: thrust, throttle, bank angle, and flight-path angle.

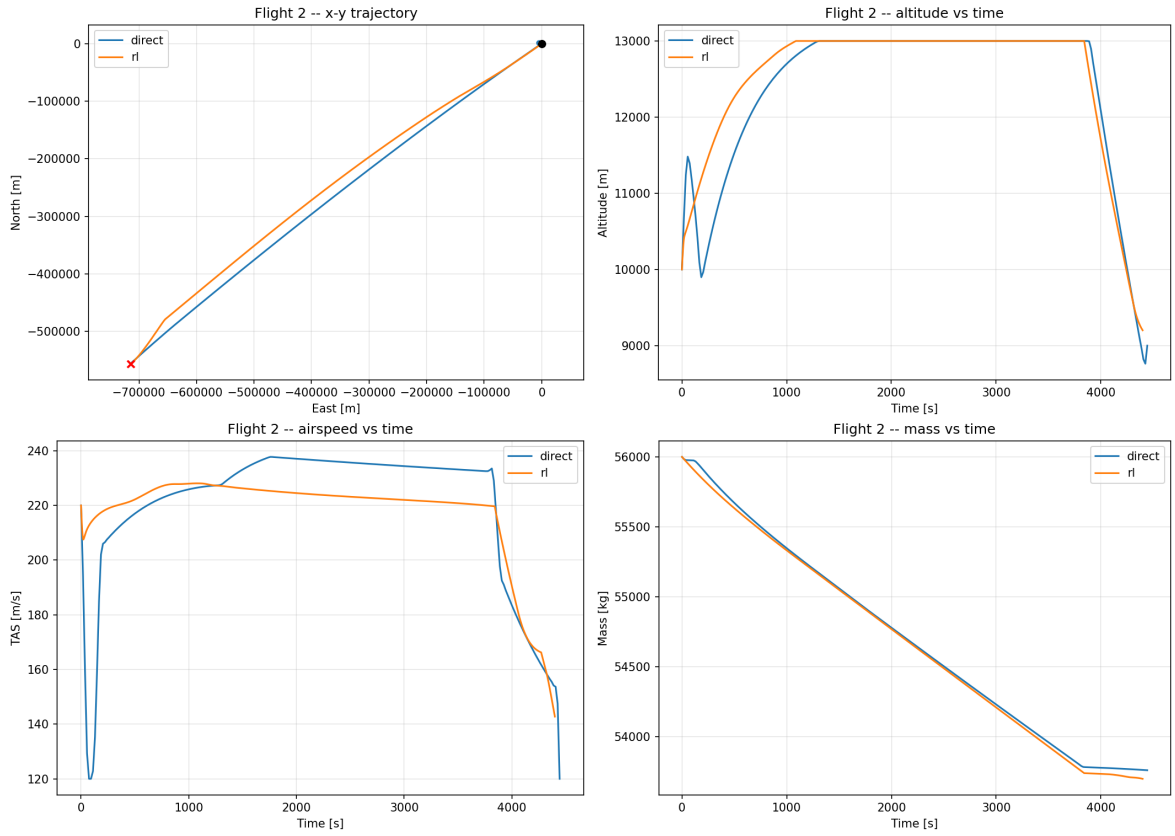


Figure 3: Flight 2 state plots: ground track, altitude, true airspeed, and mass.

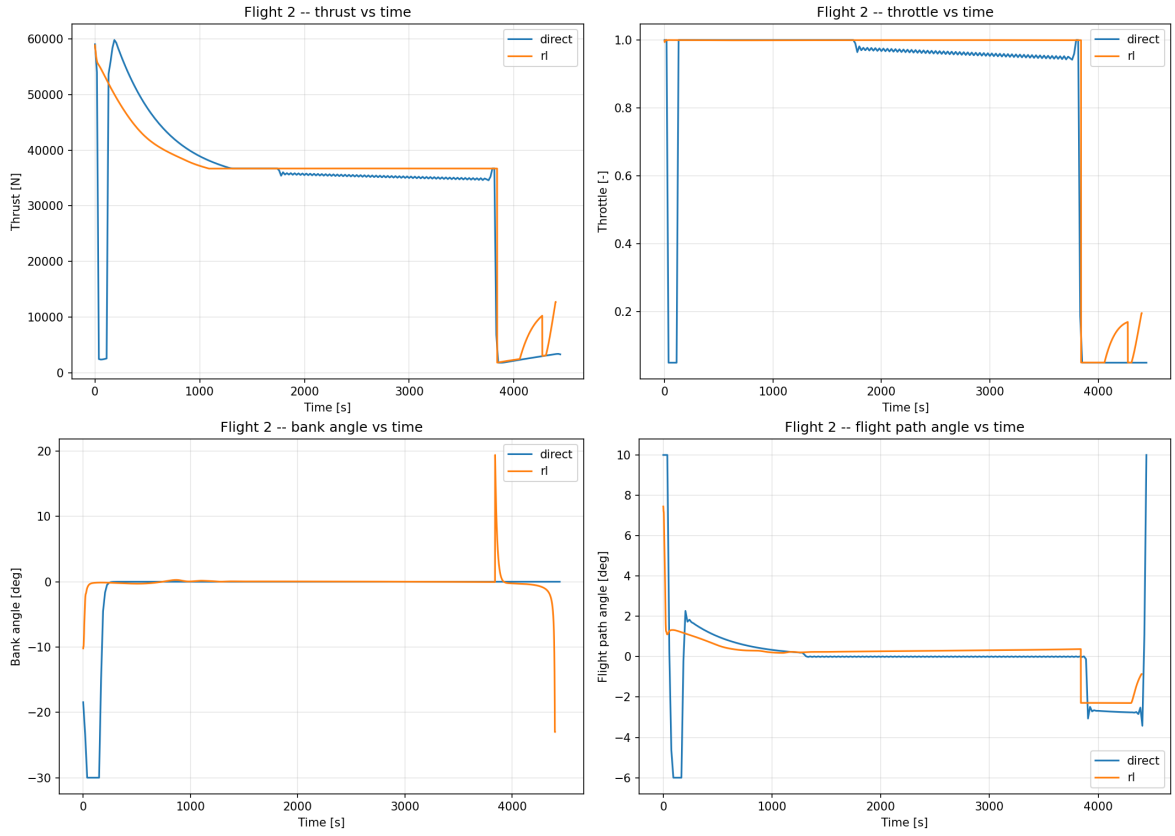


Figure 4: Flight 2 control plots: thrust, throttle, bank angle, and flight-path angle.

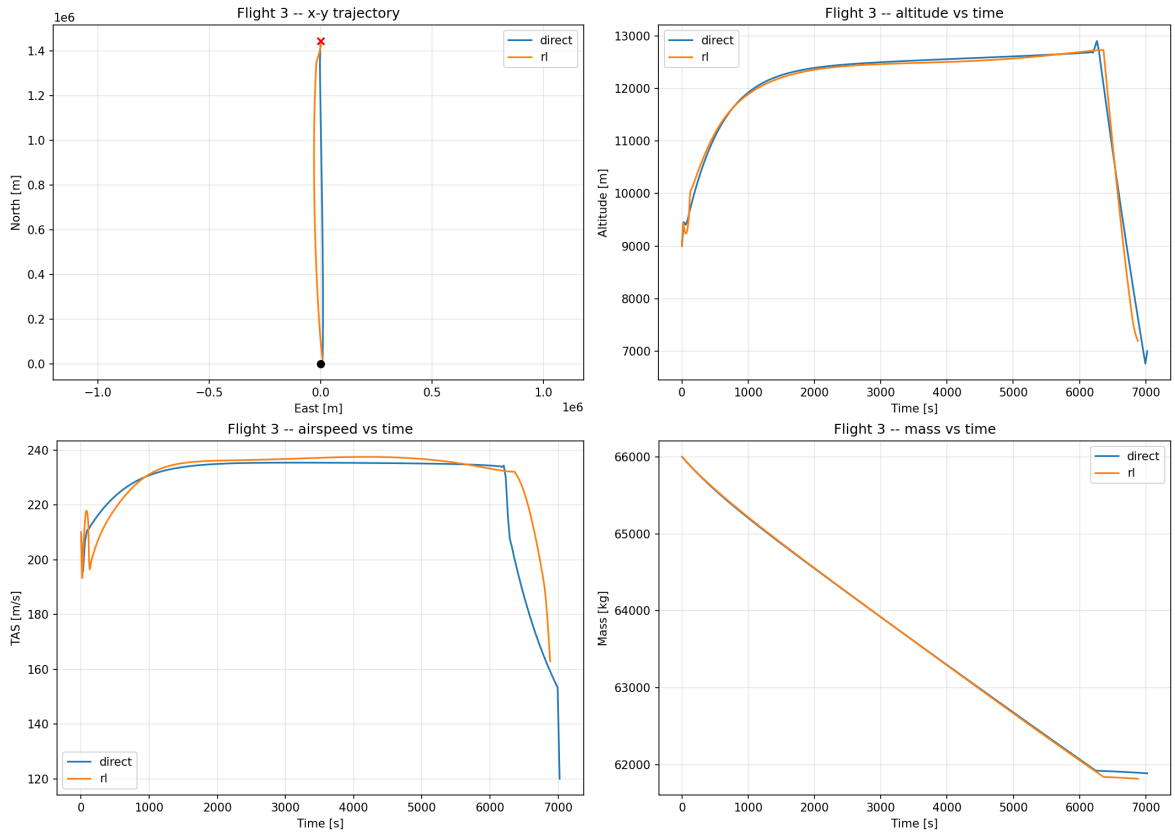


Figure 5: Flight 3 state plots: ground track, altitude, true airspeed, and mass.

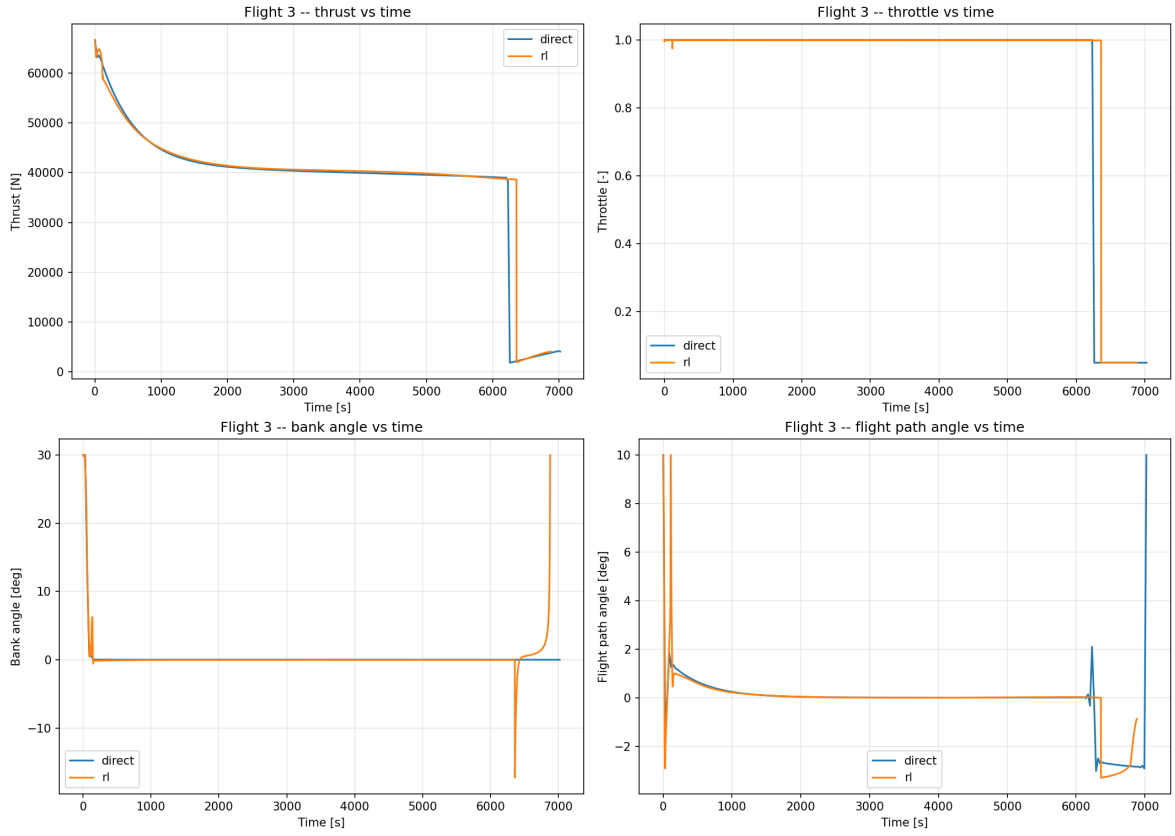


Figure 6: Flight 3 control plots: thrust, throttle, bank angle, and flight-path angle.

7 Conclusions

The direct collocation method is the strongest exact optimizer: it satisfies the terminal waypoint exactly and gives the reference objective values. Two engineering details that turned out to be necessary in practice are documented above: (i) enabling IPOPT's acceptable-termination thresholds so that the Flight-2 cold solve can exit cleanly at its true minimum rather than chase tol $= 10^{-6}$ on an already-feasible iterate, and (ii) keeping both the cold start and the Flight-1 warm start as parallel candidates and selecting on lowest cost — for all three flights, the cold-start solution beats the warm-start local optimum, sometimes by 8–19% in fuel. The closed-loop policy method is less exact but operationally useful: a single unified BC checkpoint trained on the multi-flight expert manifold reaches the 500 m terminal disk on all three nominal flights, with fuel +4.6%, +2.7%, and +1.7% above the direct reference on F1, F2, and F3 respectively. The policy's main limitation is that it satisfies a finite arrival disk rather than exact endpoint equality, which is why the direct method remains the benchmark for the optimal-control cost. Out-of-training robustness is moderate on Flights 2 and 3 (60% disk reach on perturbed Gaussian ICs) but weak on the longer Flight 1 (15%), indicating that on long horizons the lack of an outer-loop track controller is the dominant source of error rather than the policy's local feedback behaviour.

A Reproduction

The main commands used to reproduce the submitted artifacts are:

```
python -m src.direct.solve --flight all --warm-start
python -m src.rl.pipeline_unified
python -m src.rl.evaluate --flight all
python -m src.rl.robustness --flight all --n-trials 20
python -m src.compare.make_report_figs
```

Code quality was checked with:

```
flake8 src/ tests/ --max-line-length=100 --extend-ignore=E203,W503
mypy src/ --ignore-missing-imports
pytest tests/ -v
```

References

[1] S. Ross, G. J. Gordon, J. A. Bagnell. "A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning." *AISTATS*, 2011.